AeroSentinel

Active Predictive Gas Isolation Matrix

Problem Statement

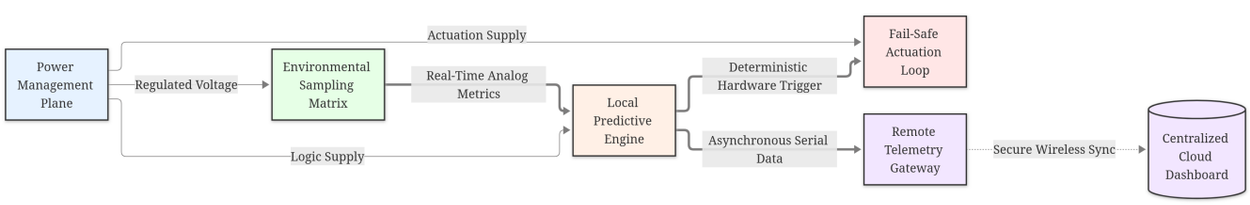
Gas leaks in homes and industrial facilities can escalate into fires, explosions, and toxic exposure. Conventional detectors are primarily reactive, triggering alarms only after gas concentration reaches a predefined threshold. They can also suffer from spatial blindness because gas distribution varies with density, buoyancy, ventilation, and airflow.

Proposed Innovation

AeroSentinel is built to actively prevent disasters instead of just passively sounding an alarm. Our system solves the problem of spatial blindness by using a custom dual-intake aspiration manifold to aspirate air from both the floor and the ceiling simultaneously. By combining this dual-intake manifold with targeted sensor fusion, the device continuously monitors hazardous combustible and volatile-gas signatures across the vertical air column, enabling earlier recognition of developing threats.

Methodology & Execution

Instead of relying solely on a fixed concentration threshold, AeroSentinel continuously establishes the normal environmental baseline and evaluates the rate at which gas concentration changes in real time. When a sustained rising threat is identified, the system autonomously initiates a local hardware response. It can simultaneously activate connected mitigation measures and isolate the gas source, transforming detection into immediate intervention. A direct manual override allows human operators to retain physical control whenever required. Remote telemetry provides live system updates through a cloud dashboard; however, the core safety response operates locally and does not depend on Wi-Fi or cloud connectivity.



Real-World Impact

AeroSentinel is designed to provide an additional layer of protection for domestic environments, small-scale facilities, and larger industrial deployments by enabling intervention before a developing gas hazard reaches a critical state. Its active response architecture can reduce dependence on human reaction time by initiating predefined mitigation measures immediately after a sustained threat is identified. Local operation also maintains essential safety functionality during network failures, making connectivity a monitoring feature rather than a safety dependency.

The system emphasizes cost-effectiveness by combining accessible sensing, embedded processing, and modular hardware to deliver advanced gas-safety capabilities without relying on expensive centralized infrastructure. Unlike conventional detectors, AeroSentinel integrates predictive analysis, multi-zone sensing, and autonomous mitigation within a single system. Its modular architecture allows additional sentinel nodes to be deployed according to the size and requirements of a facility, making the approach adaptable from small environments to larger installations. This provides a practical pathway for upgrading conventional gas detection toward more proactive and scalable safety infrastructure.